

JustEnough

Prove you qualify.
Reveal nothing else.
Claim only once.

Zero-knowledge eligibility and one-per-person claims for grants, scholarships and airdrops, on Midnight.

Code: <https://github.com/Ryugi62/justenough>

Grant application

Full name

Date of birth

Home address

Student ID

✓ Eligible · first claim

What the program learns

PROBLEM

Public blockchains force a bad trade-off: expose your users, or get farmed by bots

1 Put it on-chain

Credentials or their hashes are public forever and link a person across every app that checks them.

Leaks identity

2 Trust a KYC server

A honeypot of ID documents and a gatekeeper everyone must trust. Not decentralized.

Single point of failure

3 Skip the check

Nothing stops one person from claiming with a thousand wallets.

Gets farmed

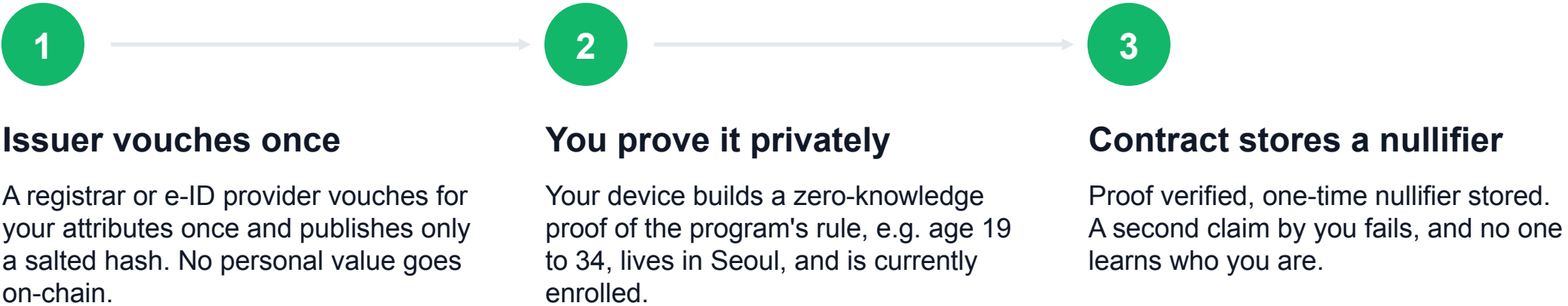
803,093

addresses flagged as potential sybils in LayerZero's 2024 airdrop, about 13% of roughly 6 million eligible, and found only after the fact by clustering wallets in public.

Source: Cointelegraph / The Defiant, May 2024

SOLUTION

Your credential stays on your device. From your claim the chain learns two facts: eligible, and first claim.



Stays on your device

Birth date, district	
Student, income	
Secret key and salt	

Visible on the public ledger

Program rule (anyone can audit it)

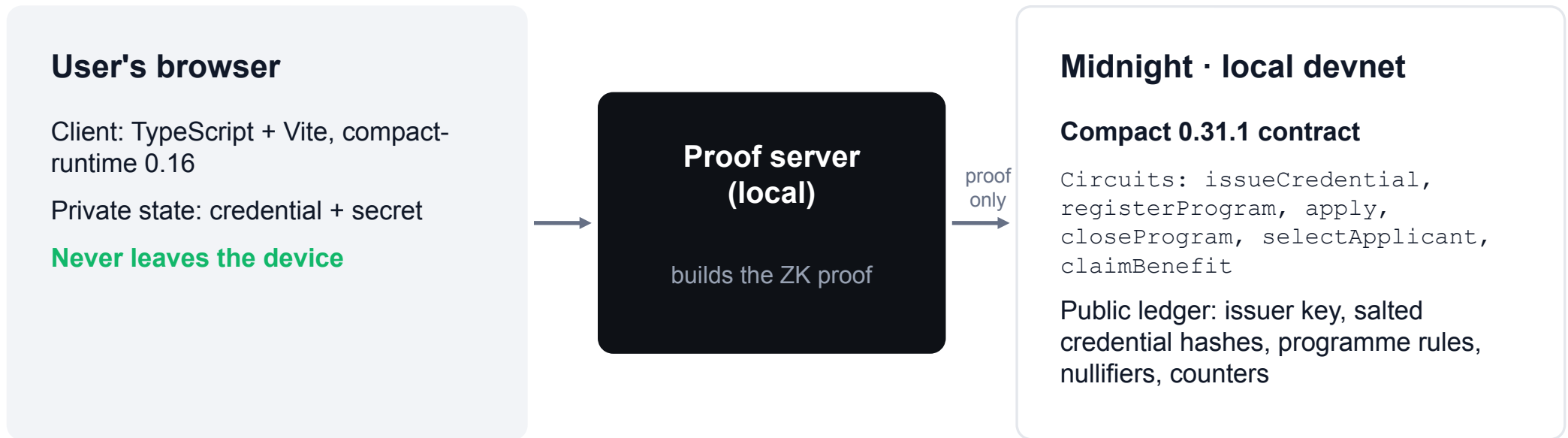
Issuer's salted hashes (which one is yours stays hidden)

Proof verified ✓

Nullifier 8332...5286 (devnet run)

TECHNOLOGY

Built on Midnight: private state stays in the browser, public state goes on-chain



Tests

112 tests passing

Testing stack

**Vitest on the compiled contract,
Playwright, GitHub Actions**

Code

<https://github.com/Ryugi62/justenough>

Not another credential-hash-on-chain project: public rules, private data

	Credential hash on-chain	KYC server + whitelist	JustEnough
Claim reveals which credential is yours	Yes, the hash itself	No	No, membership proof
Trusted server needed at claim time	No	Yes	No
Blocks one person claiming twice	Not by itself	Yes, by collecting IDs	Yes, by nullifier
Claims linkable across programs	By anyone (same hash)	By the server	No
Eligibility rules auditable by anyone	Rarely	No	Yes, on-chain

The rules are public on-chain, so a program cannot quietly change who qualifies.

Grants, scholarships and airdrops can stop collecting ID documents

0

ID documents uploaded to the program

1

nullifier stored per claim, nothing else about you

3

yes/no checks replace a full ID document in our example grant

A

Applicants

Stop uploading ID documents to every program they apply to.

P

Programs, grant DAOs, airdrops

No personal-data liability, and double claims are blocked by construction, not by after-the-fact wallet clustering.

G

Public agencies

The same flow fits scholarship and youth-grant eligibility checks.

FUTURE SCOPE

Next: real issuers, revocation and a mobile wallet

Now: hackathon MVP

- Demo issuer account
- 112 tests passing
- End-to-end flow, three roles

Next

- University registrar and e-ID issuers
- Revocation lists (expiry already enforced)
- Independent audit of the circuits

Later

- Mobile wallet with QR hand-off
- Richer rules: set membership, OR-conditions
- Unlinkable reputation across programs

Code: <https://github.com/Ryugi62/justenough>

Built from scratch starting 2026-09-24, inside the 3rd-Web-Hack submission period.